
Correlated Failure as an Indirect Signature of Shared Model Provenance: A Population-Scale Complement to Instance-Level Attribution

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Abstract

1 Contributive and corroborative attribution answer “which training data explains this
2 output,” one output at a time, and need access to training data or model internals
3 that is routinely unavailable for open leaderboard submissions. This idea-track
4 paper describes a training-data-free, population-scale signal available even when
5 neither is. After conditioning on the exact, fit-free null implied by Rasch sufficiency,
6 1,228–1,362 open models per benchmark share residual failure correlation 2.9–
7 11.9× its null, concentrated in 4–12 latent dimensions rather than one shared factor,
8 and the signal survives removing 21–64% of models as near-duplicates. We tried
9 to decompose it against declared lineage and could not: a pre-registered gate found
10 a declared parent model for only 23.3% of models, below the 40% threshold set
11 in advance, so the analysis was dropped rather than run on a biased subset. We
12 think that failure is itself the useful datum for this venue: it is a concrete, quantified
13 instance of the provenance-opacity problem this community studies, and we pose
14 the residual correlation matrix, together with the coarse family labels we do have,
15 as a candidate for what an aggregate, black-box provenance signal could look like
16 when nothing about training data is disclosed.

17 1 The gap we are pointing at

18 Contributive attribution estimates the causal effect of specific training examples on a specific output
19 (Koh & Liang, 2017; Park et al., 2023); corroborative attribution finds sources that support a specific
20 output (Worledge et al., 2024). Both operate at the level of one model and one output. Neither is
21 designed for the regime of an open leaderboard with thousands of independently submitted models,
22 most of undisclosed lineage, where the question is not “which example explains this generation” but
23 “which models share unstated ancestry with which others, and can that be seen at all without the
24 training data.”

25 We met this question from the measurement side, while characterising correlated failure between
26 open language models on standard multiple-choice benchmarks. We report what we found, and what
27 we could not do, because the gap between what we needed and what exists seems to us a useful data
28 point for this workshop even though we do not close it.

29 2 A population-scale, training-data-free residual signal

30 Let $F_{im} = 1$ iff model i fails item m , with c_m the number of models failing item m .

31 **Lemma 1 (Mean co-failure depends only on item margins)** $O = \frac{1}{MN(N-1)} \sum_m (c_m^2 - c_m)$;
 32 *hence any randomization preserving the item margins leaves the mean pairwise co-failure rate*
 33 *O exactly invariant, with zero variance.*

34 The row and column margins of F are the jointly sufficient statistics of the Rasch model, so condition-
 35 ing on both, sampled with curveball (Strona et al., 2014) and verified against exhaustive enumeration
 36 on small fibres, gives an exact, fit-free null under which mean-level co-failure is provably uninfor-
 37 mative about shared behaviour (Lemma 1; the same degeneracy is Schluter’s V-ratio in ecology and
 38 Cronbach’s α in psychometrics (Schluter, 1984; Gotelli, 2000; Cronbach, 1951)). At that null, on
 39 the archived Open LLM Leaderboard (five benchmarks, 1,228–1,362 models each, harvested from
 40 result files at zero inference cost), the residual correlation that remains after the difficulty degeneracy
 41 is removed is $2.9\text{--}11.9\times$ its null and spectrally concentrated: on ARC-Challenge, six eigenvalues
 42 exceed the null’s spectral edge and the leading eigenvalue carries 54% of squared residual mass, the
 43 signature of several latent dimensions rather than one shared factor. This is not a duplicate-submission
 44 artifact. Clustering models by pairwise agreement and keeping one representative per cluster at
 45 threshold 0.90 removes 900 of 1,362 ARC models and moves the participation ratio of the residual
 46 spectrum from 23.6 only to 29.0 (its exact-margin null falls from 449.6 to 289.3); recomputing
 47 every statistic at threshold 0.95, which discards 21–64% of each benchmark’s models with the Rasch
 48 fit and the null redrawn, moves the rms excess ratio by at most $1.14\times$ on every benchmark (ARC
 49 $5.58 \rightarrow 5.44$). Whatever produces this structure survives the crudest form of provenance filtering
 50 available to us.

51 That last clause is doing real work: it is the crudest form available because we do not have a better
 52 one. We do not observe pretraining corpora, fine-tuning data, or distillation relationships for the large
 53 majority of these models. Two attribution routes were tried. The leaderboard’s optional, self-reported
 54 `base_model` field resolved a declared parent for 317 of 1,362 ARC models (23.3%), with 315 model
 55 repositories no longer retrievable at all; we had pre-registered a 40% coverage threshold below
 56 which a lineage decomposition would not be trusted, so it was dropped rather than reported on a
 57 biased subset. Name-string matching against known family names attributes 34.5–35.7% of model
 58 identifiers per benchmark, with roughly 125 models per benchmark carrying merge-related names
 59 and the five largest named families (Mistral, Llama-2, Llama-3, Mixtral, and a fifth that varies)
 60 covering only 25–26%. We are not claiming the residual correlation *is* shared provenance; the honest
 61 statement is narrower and, we think, more useful to this audience: standard attribution machinery
 62 cannot currently tell us either way, because it needs exactly the access this population does not
 63 provide.

64 3 Why this might be attribution’s problem, not measurement’s

65 Three properties of the residual structure make it look more like a provenance signal than a behavioural
 66 coincidence, and we label the inference that follows as a conjecture. First, it is not explained by
 67 discrimination heterogeneity: a two-parameter item-response simulation with $\text{sd}(\log a) = 0.60$ yields
 68 rms residual correlation 0.076 against an observed 0.248 on ARC. Second, it is not explained by
 69 literal duplicate submissions, as above. Third, its dimensionality is benchmark-specific and broadly
 70 stable under deduplication (4–6 dimensions above the null’s spectral edge on four benchmarks, 12 on
 71 HellaSwag). *Conjecture*: this is the structure one would expect from a handful of dominant pretraining
 72 lineages or shared distillation teachers producing correlated errors along a few behavioural axes,
 73 rather than from one dominant lineage or from fully independent training runs. None of this is proof,
 74 and our own simulations show that a population with two ability dimensions and no shared lineage
 75 reproduces much of the effect ($\lambda_1^2 / \sum \lambda_k^2 = 0.457$ against the observed 0.540), so unmodelled ability
 76 dimensions remain an alternative we cannot exclude. What we can say is that the exact conditional
 77 null gives a population-scale test that is invariant to item difficulty by construction, costs no training-
 78 data access, and is sensitive to structure at a scale (thousands of models) at which per-example causal
 79 attribution is not, to our knowledge, currently run.

80 4 An idea-track question

81 We pose this as a question rather than a method, and it is now concretely testable with the coarse
 82 labels above: do the leading residual eigenvectors of a margin-conditioned co-failure matrix separate,

even loosely, the Mistral-7B-derived population from the Llama-2-derived one, or the merge-named models from the rest? If they do, the residual matrix would be a training-data-free, black-box complement to contributive and corroborative attribution, useful precisely in the regime where those methods’ preconditions (access to training data; an identified individual output to explain) fail, such as a regulatory or journalistic audit of a leaderboard’s actual diversity of origin. If they do not, and the residual structure is better explained by convergent training choices than by shared lineage, that is itself a useful negative result about how much population-scale behavioural correlation should ever be read as a provenance signal. We do not know which is true; we have not run the eigenvector-versus-family comparison, and we think this venue is the right place to decide whether it is worth running.

5 Limitations

This paper reports a measurement and a gap, not an attribution method; we performed no causal or corroborative attribution. Discrimination heterogeneity is ruled out as a sufficient explanation but unmodelled ability dimensions are not. The name-string family labels are a lower bound on lineage, not ground truth. Claims are restricted to the evaluated open-model ecosystem on multiple-choice benchmarks. Code, data, and the pre-registration with dated amendments will be linked in the camera-ready. A large language model was used as a coding and drafting assistant; the authors specified and pre-registered the hypotheses, verified every derivation and number against executed-run artifacts, and are responsible for the content.

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